

Transmission Media

Guided Media

- (1) Twisted pair cable
- (2) Coaxial cable
- (3) Fiber optic cable

UnGuided Media

- (1) Radio waves
- (2) Microwaves
- (3) Infrared

Switching

1) Circuit Switch (physical link $\rightarrow$ dedicate channel)

Phase circuit switch

- ✓ Connection setup
- ✓ Data Transfer
- ✓ Connection Teardown

Efficiency Circuit switch (less efficiency)

Delay in Circuit Switch (Not delayed)

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(2) Packet switching High efficiency $\rightarrow$ Data divide packet $\rightarrow$ high

Two type Packet Switching

(1) Datagram Networks (Independently) ^{packet refer}
_{Datagram}

(2) Virtual Networks (logical path)

(3) Message Switching (less use)

* Time Division Switch Inside switch

Data Link Layer (node to node)

Service data link layer

* Four service provide data link layer

(1) Framing

(2) Flow control

(3) Error control

(4) Congestion control

Two categories link

Point to point link

Broadcast link

Two Sublayers of Data link layer

(1) Data Link control (DLC)

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Why link layer addressing?

IP address are the identifiers at the network layer.

imp Three Types of address

- (1) Unicast
- (2) Multicast
- (3) Broadcast

Address Resolution Protocol (ARP)

Types of Error

- (1) Single bit error
- (2) Burst bit error

Redundancy (main concept error correction and detection → adding extra bit)

Error Detection

Error Correction

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Coding

2 Types of coding

(1) Block coding $n = kt + r$

(2) Convolution coding *prime!*

Hamming Distance (number of size same but ~~mean~~ diff)

Minimum hamming distance (Use detect error)

$$d_{\min} = s + 1$$

Linear block code

Parity check code $n = k + 1$ (common error detecting code)

Cyclic codes

Cyclic Redundancy Check (CRC) Subset cyclic codes

CRC is used in network such as LAN and WAN.

CHECKSUM (error detection technique)

Checksum mostly used network and transport layer rather than data link layer.

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Forward ERROR Correction

Using Hamming Distance

$$d_{\min} = e + 1 \quad e = 2t$$

$$d_{\min} = 2t + 1$$

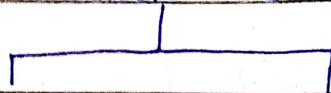
Chunk Interleaving

DATA Link control (DLC) Services

- (i) Framing
- (ii) Flow Control
- (iii) Error Control

Frame Type

- (1) Fixed ^{frame} size
- (2) Variable ^{frame} size



Connection oriented	Bit oriented
Framing (byte oriented)	Framing

Flow control

Error control → Two ways → Acknowledged (ACK), Without ACK

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Data link layer Protocol

four protocol data link layer deal with flow control and error control.

^{Simple}
(1) ~~STOP~~ Protocol (2) STOP and WAIT Protocol

~~Quater State Machine~~

(3) Go-Back-N Protocol (4) Selective-Report Protocol

Simple ~~STOP~~ Protocol → neither flow nor error control

Stop and wait control → use both flow control and error control

Piggybacking

both simple and stop-and-wait protocols are design for unidirectional communication.

High-Level Data Link Control (HDLC)

HDLC is bit oriented data link layer use for:
Point to Point.

Multipoint

Configuration & Transfer Modes in HDLC

HDLC provide two common transfer mode that can be use in different configurati

Normal Response Mode (NRM)

Asynchronous Balance Mode (ABM)

Framing

HDLC define three type of frames:

information frames (I-frames)

supervisory frames (S-frames)

Unnumbered frames (U-frames)

Point to Point (Protocol) (PPP)

Multiplexing in PPP

Three set of protocol

Link control protocol

Authentication protocol

Network Control Protocol (NCP)

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Authentication Protocol

PPP Authentication

Password Authentication
Protocol (PAP)

Challenge Handshake
Authentication Protocol (CHAP)

Media Access Control (MAC)

Random Access

ALOHA earliest random access method was developed in early 1970.

Slotted ALOHA Slotted Aloha was invented to improve the efficiency of pure Aloha-

CSMA carrier sense multiple access
Sense before-transmit

CSMA/CS carrier sense multiple access / collision avoided

avoid collision $\rightarrow$ used in wireless LAN



through use of strategies

(i) Interframe space (IFS)

(ii) The contention Window

(iii) Acknowledgement

Controlled Access cannot send unless

authorized by other station

We discussed three controlled access method

(1) Reservation

(2) Polling

(3) Token passing

Channelization (Chanal Partition)

We discuss three protocol:-

Frequency division Protocol

Time division Protocol

Code division Protocol

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Ethernet ^(Physical Layer) Protocol most commonly use LAN
ethernet can be used wired or wireless protocol

IEEE Project 802 started 1985- function of
physical layer and data link layer of major LAN protocol

Ethernet evolution developed 1970.

Since it has gone through four generation

- (1) Standard ethernet (10 Mbps) speed
- (2) Fast ethernet (100 Mbps) speed
- (3) Gigabit ethernet (1 Gbps (1000 Mbps))
- (4) 10 Gigabit Ethernet (10 Gbps)

Addressing in Standard Ethernet

each device on an ethernet network has a

MAC network interface card (NIC)

MAC provide link layer address called media access control (MAC)

MAC Ethernet uses a 48-bit address (6 bytes)

It written hexadecimal → separated by colon-

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Access Method in Standard Ethernet

The standard Ethernet chose CSMA/CD with
1-Persistent Method.

Efficiency of Standard Ethernet

$$\text{Efficiency} = \frac{1}{1 + 6.4a}$$

a = number of frames that can fit on medium

Change in Standards

Some change were made in 10-Mbps
because of these changes ethernet involved:

- ✓ Bridged Ethernet
- ✓ Switched Ethernet
- ✓ Full-Duplex Ethernet

Implementation

10 Gigabit Ethernet network only in full-duplex mode.

Since there is no share connection.

CSMA/CD is not used.

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Other wired network

LAN, WAN

Network that connect a LAN to an ISP.

Internet Service Provider

Telephone Network

The telephone network started in the late 1800s.

Plain old telephone system (POTS) was originally analog system. used analog system transmit voice only.

In 1980 with rise computer era network-

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Access Network Connect LAN to ISP

WAN Transfer data over long distance

POTS (Plain old Telephone system) old analog telephone system

Modern Telephone Network Both digital and analog
supports voice and data

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LOCAL ACCESS TRANSPORT AREAS (LATAS)

A LATA can be small or large metropolitan area -

A small state may have a single LATA.

A large state may have several LATAs.

Intra LATA and Inter LATA Services

Service offered by telephone companies inside LATA are called Intra LATA service.

IMP

Local exchange carrier (LEC)

Carrier that handles Intra LATA are called LEC.

Interexchange Carriers (IXCs)

Signalling

In the network the telephone network used circuit-switched communication.

Signal system became automatic.

As telephone signal more complex and large thus the signal more advance.

Service

Telephone companies provide two types of service.

✓ Analog Service

✓ Digital Service

Digital Subscriber Line (DSL)

DSL provide high speed.

Type of DSL

(1) ADSL Asymmetric DSL

(4) SDSL Symmetric DSL

(2) VDSL Very high bitrate DSL

Traditional Cable Networks

Cable TV started in late 1940s.

Known as CATV (Community Antenna Television)

A large Antenna was placed on hill or tall building.

Hybrid Fiber Coaxial (HFC) Network

Second generation of cable network is called HFC

HFC two types -

Fiber optic cable

Coaxial cable

Synchronous Optical Network (SONET)

Sonet is Wide area network.

Internet version of sonet is called SDH (Synchronous digital Hierarchy) standardized by ITU-T

Architecture sonet three parts

(1) Signal (2) Device (3) Connections

SONET Architecture

(1) Signal

- ✓ Synchronous Transport Signals (STS)
- ✓ Optical carriers (OCs)
- ✓ Synchronous Transport Module (STM)

(2) Devices

- ✓ STS Mux/Demux
- ✓ Regenerators
- ✓ Add-Drop Multiplexer and Terminal

(3) Connections

- ✓ Section
- ✓ line
- ✓ path

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Sonet layers

four layers

✓ The Path layer

✓ The Line layer

✓ The section layer

✓ The photonic layer

The layer correspond to both physical and data-link layer

SONET frames

Each STS signal send 8000 frame per second.

Each frame is matrix per second.

STS-1 Frame:

9 rows X 90 columns

Total = 810 bytes

STS-3 Frame:

9 rows X 270 columns

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ATM (Asynchronous Transfer Mode) high speed

Architecture

ATM is cell-switch network.

The switches are connected network to network interface (NNI)

Wireless Communication

Wireless communication is fast growing technology

There is huge demand for connecting device without cables.

Wireless WAN widely used in

- College Campus

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Wireless WAN widely used in

- College Campus
- Offices
- Public areas

Characteristic Wireless LAN

- ✓ Attenuation
- ✓ Interference
- ✓ Multiple Propagation
- ✓ Error

Access Control

- How a wireless device can access the shared medium (air)
- CSMA/CD (used in wire LAN) does not work for three reasons

(1) Wireless host cannot send and receive at same time.

(2) Hidden station Problem

(3) Large distance

IEEE 802.11 Project

Wireless LAN called IEEE 802.11 (physical + Data link)

WIFI certified version IEEE 802.11.

Architecture

Two kind of service

Basic Service Set (BSS) small LAN

Extended Service Set (ESS) multiple LAN

Types of station

No transition mobility

BSS - Transition mobility

ESS transition mobility

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MAC Sublayer

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IEEE 802.11 define two MAC Sublayer

The Distribution Coordination Function (DCF)

The Point Coordination Function (PCF)

IEEE 802.11 Frame types

Management Frames

Control Frames

Data Frames

Physical Layer

All physical implementation operate ISM.

ISM (Industrial, Scientific and Medical) band is unlicensed band.

ISM frequency range

902 - 928 MHz

2.400 - 4.835 GHz

5.725 - 5.850 GHz

BLUETOOTH

- Bluetooth is wireless LAN technology.
- Bluetooth LAN is an ad hoc network.
- Bluetooth devices called (budget) find each other automatically.
- They form small network called piconet.

Architecture

Bluetooth two type of network

- 1 Piconet
- 2 Scatternd

BLUETOOTH

Bluetooth technology is base on IEEE 802.15 standard.

IEEE 802.15 define wireless Personal Area Network (PAN)

PAN operate small areas

a room

a hall

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BLUETOOTH

Bluetooth device has built in short range radio transmitter.

The current data rate 1Mbps with 2.4 GHz bandwidth

IEEE 802.11b wireless LAN and bluetooth lan-

BLUETOOTH Layers

Bluetooth use multiple physical layer

layer are design special short range wireless communication-

Connecting Devices

Three kind of connecting devices-

✓ Hubs

• • • • • Inner Switches

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Connecting Devices

Three kind of connecting devices-

- ✓ Hubs

- ✓ Link layer Switches

- ✓ Routers

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Switch vs Hub

Feature	Hub	Switch
Layer	Physical	Physical + Data link
Address checking	No	Yes (MAC address)
Filtering	No	Yes
Collisions	Many	Eliminated (No)
Carrier Sensing	Require	Not require
Device Type	Same Device	Can connect heterogeneous device

~~Switch~~ Router vs Switch

Feature	Switch	Router
OSI / Internet layer	Physical + Data link layer	Physical + Data link + Network
Type of address use	MAC address (physical)	MAC + IP address (Physical + logical)
Main Function	Connect device within the same LAN	Connect different network
Address change	Does not change MAC address	Changes source and destination MAC address
Collision	Eliminate collision	Eliminate collisions

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Speed	Very fast	Slightly slower
Use case	LAN, office, campus	Internet access, WAN inter-networking
Loop Problem	Loop can occur (solved by spanning Tree)	No Loop Problem

Virtual LAN (VLAN)

Virtual line is a logical (virtual) LAN created by software not physical wiring.

Membership is Geographical

Characteristic of Virtual LAN

VLAN can be based on

Interface number

Port number

MAC address

IP address

IP multicast address

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Combination of above

Configuration of VLAN

Three way

Manually

Semi-Automatically

Automatically

Advantage VLAN

Cost and Time Reduction

Creating virtual Workgroups

Security

Modern Access Technology

Teleco (Telecommunication)

HFC Hybrid fiber coaxial

FTTx Fiber tex transformation

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Feature	FTTC Fiber to the Curb	FTTH Fiber to the Home
Fiber reach	Partial	Full
Last Segment	Copper/coax	Fiber
Speed	High	Very High
Future-proof	Limited	Excellent
Cost	Lower	Higher

ARP Address Resolution Protocol

CRC Cyclic Redundancy Check

DLC Data link control

FSM Finite state Machine

ACK Acknowledge

HDLC High Data Link Control

PPP Point to Point

NRM Normal Response Mode

ABM Asynchronous Balanced Mode

I-Frame information frames

S-frames Supervisory frames

U-frames Unnumbered frames

LCP Link control Protocol

APs Authentication Protocol

NCPs Several Network control protocol

PAP Password Authentication Protocol

CHAP Challenge Handshake Authentication Protocol

MAC Media Access Control

CSMA Carrier Sense Multiple Access

CSMA/CD Carrier Sense Multiple Access/Collision Detection

CSMA/CA Carrier Sense Multiple Access/Collision Avoidance

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FDMA Frequency Division Multiple Access

TDMA Time Division Multiple Access

CDMA Code Division Multiple Access

IEEE

NIC Network Interface Card

MAC Media Access Control

ISP Internet Service Provider

POTS Plain Old Telephone System

LATAS Local Access Transport Areas

LEC Local Exchange Carrier

IXCs Interexchange Carriers

DSL Digital Subscriber Line

ADSL Asymmetric DSL

VDSL Very-high DSL

HDSL High DSL

SDSL Symmetric DSL

CATV Community Antenna Television

HFC Hybrid Fiber Coaxial

UTP Unshielded Twisted Pair

SONET Synchronous Optical Network

SDH Synchronous Digital Hierarchy

STS Synchronous Transport Module

OCs Optical Carriers

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STM Synchronous Transport Module

ATM Ayschronous Transfer Mode

UNI User to Network Interface

NNI_s Network to Network Interface

BSS Basic Service Set

ESS Extended Service Set

DCF Distributed Coordination Function

PCF Point Coordination Function

ISM Industrial Scientific and Medical

PAN Personal Area Network

VLAN Virtual local Area Network.

Telco Telecommunication

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